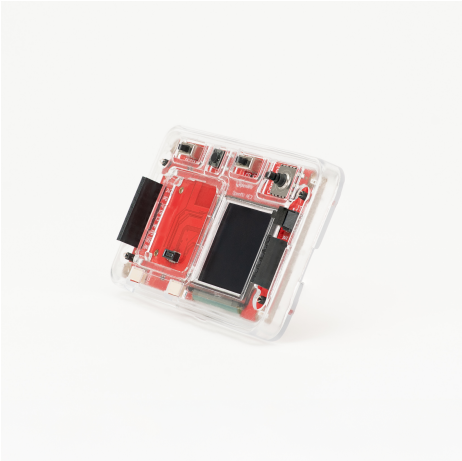
OpenMV® AE3 OLED Breakout

Datasheet

SKU: OMV-AE3-OLED-BREAKOUT

Description

The AE3 OLED Breakout is an expansion board providing full access to the OpenMV AE3's capabilities. It features a 128x128 RGB OLED display with a software-controllable on/off switch, a 5-direction joystick with center button for user interaction, two QWIIC connectors for I2C peripheral daisy-chaining, and all 10 AE3 GPIO pins broken out to 3.3 V headers. For development and debugging, the board includes an ARM 10-pin JTAG header for SWD debugging (SEGGER J-Link compatible), four ground hooks for oscilloscope probing, a hardware reset button, and a recovery switch for secure enclave UART access.



Target Areas

Development, prototyping, debugging, HMI, IoT dashboards, sensor integration, interactive demos, field testing.

Key Features

- **128x128 RGB OLED** — On-board color OLED display with software-controllable on/off switch to save power when not needed.
- **Joystick + button** — 5-direction joystick (up/down/left/right) with center press button for menu navigation and user input.
- **Full GPIO breakout** — All 10 AE3 GPIO pins broken out to 3.3 V pin headers for easy prototyping and wiring.
- **Dual QWIIC** — Two QWIIC I2C connectors for daisy-chaining peripherals. Compatible with Seeed Grove and Adafruit STEMMA.
- **JTAG debug** — ARM 10-pin JTAG header for SWD debugging with SEGGER J-Link and compatible debuggers.
- **Ground hooks** — Four ground hooks for oscilloscope probing and test equipment attachment.
- **Recovery mode** — On-board recovery switch for secure enclave UART access and firmware recovery.

Applications

The AE3 OLED Breakout is ideal for development, prototyping, debugging, HMI interfaces, IoT dashboards, sensor integration, interactive demos, and field testing of OpenMV AE3 applications.

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1 Features and System Components

1.1 Display

Component	Details
OLED Screen	128x128 pixel RGB OLED display. Full-color graphics and text rendering. Can be disabled via on-board switch to save power or free up SPI pins.

1.2 User Input

Component	Details
Joystick	5-direction joystick (up, down, left, right) with center press button. Directly readable from MicroPython for menu navigation and interactive applications.
Reset Button	Hardware reset button for the AE3 module.
Recovery Switch	Dedicated switch for secure enclave UART access. Enables firmware recovery and secure boot configuration.

1.3 Connectors

Component	Details
GPIO Headers	All 10 AE3 GPIO pins broken out to 0.1-inch pin headers at 3.3 V logic levels.
QWIIC / I2C	2 QWIIC I2C connectors for daisy-chaining peripherals. Compatible with SparkFun QWIIC, Seeed Grove, and Adafruit STEMMA ecosystems.
JTAG	ARM 10-pin JTAG header for SWD debugging. SEGGER J-Link compatible. Enables full source-level debugging of AE3 firmware.
Ground Hooks	4 ground hooks for oscilloscope probing, logic analyzer attachment, and general test equipment.

1.4 Compatibility

Component	Details
OpenMV AE3	Designed exclusively for the OpenMV AE3 camera module. The AE3 plugs directly into the breakout board via its board-to-board connector.

2 Pinout

2.1 SPI OLED Pins (ER-OLED015-1C)

Pin	Function
P0	MOSI
P2	SCLK
P3	SSEL
P7	RSTN
P8	RS

2.2 I2C GPIO Expander Pins (PCA9674A)

Pin	Function
P4	SCL
P5	SDA
P9	INT

2.3 Joystick Pins (AS90R)

Pin	Function
PCA9674A P0	Right
PCA9674A P1	Up
PCA9674A P2	Left
PCA9674A P3	Down
PCA9674A P4	Center Button
PCA9674A P6	OLED Enable Switch

2.4 QWIIC Pins

Pin	Function
P4	SCL
P5	SDA

2.5 Power Consumption

Condition	Current Draw
AE3 on streaming (OLED off)	45 mA @ 5 V
AE3 and OLED on streaming	130 mA @ 5 V

3 General Info

Dimension	Value
Length	97 mm
Width	59 mm
Height	15 mm
Weight	43 g
Operating Temperature	-20°C to +70°C
Storage Temperature	-20°C to +80°C
Country of Origin	China
HS Code	8473.30.1180
UPC	602772227144
SKU	OMV-AE3-OLED-BREAKOUT
ECCN	EAR99

4 Company Information

Field	Details
Website	https://openmv.io
Documentation	https://docs.openmv.io
Support	https://forums.openmv.io

5 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/ae3-oled-breakout
3D Model	https://grabcad.com/library/openmv-ae3-oled-breakout-board-1
3D Mounts	https://grabcad.com/library/ae3-oled-breakout-mounts-1

6 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.